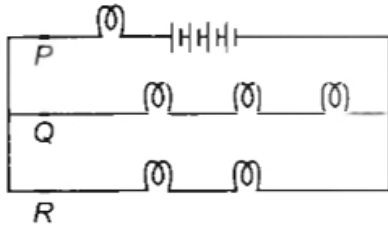




ELECTRICITY -02

Class 10 - Science

1. Which of the following statements about the current in given circuit is correct? [1]



- a) R has maximum current.
 - b) Current at Q is greater than the current at R.
 - c) P has maximum current.
 - d) Current at R is greater than the current at P.
 2. Consider the room temperature is 24°C in summer, the electrical resistance of thermocoil which is used in the AC unit is $150\ \Omega$. Then calculate the temperature of the thermocoil if the electrical resistance is $175\ \Omega$. Given the temperature coefficient of the thermocoil is $2.98 \times 10^{-4}\ ^{\circ}\text{C}^{-1}$ [1]
 - a) $583\ ^{\circ}\text{C}$
 - b) $512\ ^{\circ}\text{C}$
 - c) $597\ ^{\circ}\text{C}$
 - d) $546\ ^{\circ}\text{C}$
 3. A piece of wire of resistance R is cut into five equal parts. These parts are then connected in parallel. If the equivalent resistance of this combination is R' , then the ratio R/R' is: [1]
 - a) $1/25$
 - b) 5
 - c) 25
 - d) $1/5$
 4. Electrical resistivity of a given metallic wire depends upon [1]
 - a) its length
 - b) its shape
 - c) its thickness
 - d) nature of the material
 5. The resistivity of copper metal depends on only one of the following factors. This factor is: [1]
 - a) Thickness
 - b) Temperature
 - c) Length
 - d) Area of cross-section
 6. If the potential difference between the ends of a fixed resistor is halved, the electric power will become: [1]
 - a) half
 - b) double
 - c) one-fourth
 - d) four times
 7. If the area of cross-section of a resistance wire is halved, then its resistance becomes: [1]
 - a) 4 times
 - b) One-half

- c) 2 times d) One-fourth
8. When the area of cross-section of a conductor is doubled, its resistance becomes: [1]
a) Half b) Four times
c) Double d) One-fourth
9. The substance having infinitely high electrical resistance is called: [1]
a) Conductor b) Insulator
c) Resistor d) Superconductor
10. A cylindrical conductor of length l and uniform area of cross-section A has resistance R . Another conductor of length $2.5 l$ and resistance $0.5 R$ of the same material has area of cross-section [1]
a) $2.5 A$ b) $5 A$
c) $\frac{1}{5} A$ d) $0.5 A$
11. How much resistance does a superconductor have? [1]
a) Considerable resistance b) Low resistance
c) High resistance d) Zero resistance
12. The resistance of the conductor is R . If the length is doubled by stretching the wire, then its new resistance will be: [1]
a) R b) $4R$
c) $8R$ d) $2R$
13. The unit of resistivity is [1]
a) ohm-m b) mho
c) ohm d) ohm/m
14. Which of the following does not apply to silver? [1]
a) The movement of electrons on their outer most orbital is tightly held together. b) The resistance provided is directly proportional to its length.
c) Their resistivity is in the range $10^{-8} \Omega$ to $10^{-6} \Omega$. d) The resistance provided is inversely proportional to the area of cross section.
15. The resistivity of a metallic wire depends on: [1]
a) Its shape b) Its thickness
c) Nature of material d) Its length
16. When the diameter of a wire is doubled, its resistance becomes: [1]
a) Double b) Four times
c) One-fourth d) One-half
17. The heat produced by passing an electric current through a fixed resistor is proportional to the square of: [1]
a) time for which current is passed b) magnitude of resistance of the resistor
c) temperature of the resistor d) magnitude of current

18. A potential difference of 20V is applied across the ends of a resistance of 5 ohms. What current will flow in the resistance? [1]
- a) 20A
b) None of these
c) 5A
d) 4A
19. If the resistance of a certain copper wire is 1Ω , then the resistance of a similar nichrome wire will be about: [1]
- a) 45Ω
b) 60Ω
c) 30Ω
d) 25Ω
20. A piece of wire of resistance 20Ω is drawn out so that its length is increased to twice its original length. Resistance of the wire in the new situation: [1]
- a) 70Ω
b) 50Ω
c) None of these
d) 80Ω
21. Keeping the potential difference constant, the resistance of a circuit is doubled. The current will become: [1]
- a) Four times
b) Half
c) One-fourth
d) Double
22. Which of the following is/are good conductor(s) of electricity? [1]
- a) Mica, Quartz
b) Metals, Mica
c) Metals, Quartz
d) Metals, Rubber
23. A voltmeter and an ammeter are connected in series to an ideal cell of emf E. If the voltmeter reading is V, and the ammeter reading is I. Then [1]
- I. $V < E$
II. The voltmeter resistance is $\frac{V}{I}$
III. The potential difference across the ammeter is $(E - V)$
IV. Voltmeter resistance + ammeter resistance = $\frac{E}{I}$
- a) III and IV
b) I and II only
c) II and III only
d) I, II, III, IV
24. The resistivity of the conductor does not depend upon its length or thickness because it is the: [1]
- a) Independent to thickness
b) Independent to length
c) Nature of material
d) Depends upon altitude
25. If the diameter of a resistance wire is halved, then its resistance becomes: [1]
- a) One-fourth
b) Two times
c) Half
d) Four times
26. The resistance of the wire when the length of the wire increases two times: [1]
- a) Becomes 3 times
b) Becomes 6 times
c) Becomes 4 times
d) Becomes 2 times
27. A cylindrical conductor of length l and uniform area of cross section A has resistance R. Another conductor of length 2l and resistance R of the same material has area of cross section. [1]

- a) $3A/2$
c) $A/2$

b) $2A$
d) $3A$

28. Which of the following is the most likely temperature of the filament of an electric light bulb when it is working on the normal 220 V supply line? [1]
a) $2500^{\circ}C$
b) $1500^{\circ}C$
c) $500^{\circ}C$
d) $4500^{\circ}C$

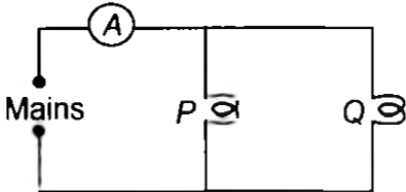
29. The resistivity does not change if [1]
a) the shape of the resistor is changed
b) the temperature is changed
c) both material and temperature are changed
d) the material is changed

30. For verifying Ohm's law, we design an electric circuit diagram in which we show the arrangement of different circuit components. We find that with respect to the resistor, the: [1]
a) ammeter and voltmeter are both connected in parallel.
b) ammeter and voltmeter are both connected in series.
c) ammeter is connected in parallel and the voltmeter in series.
d) ammeter is connected in series and the voltmeter in parallel.

31. A battery of 9V is connected in series with resistors of $0.2\ \Omega$, $0.3\ \Omega$, $0.4\ \Omega$, $0.5\ \Omega$, and $12\ \Omega$. How much current would flow through the 12 resistors? [1]
a) 0.2 amp
b) 0.61 amp
c) 0.67 amp
d) None of these

32. If 3 resistances of 3 ohm each are connected in parallel, what will be their total resistance? [1]
a) $1\ \Omega$
b) $3\ \Omega$
c) $2\ \Omega$
d) None of these

33. How will be the reading in the ammeter A affected if another identical bulb Q is connected in parallel to P? (The voltage in the mains is maintained at a constant value.) [1]



a) The reading will be reduced to one-half.
b) The reading will not be affected.
c) The reading will be increased four-fold.
d) The reading will be doubled.

34. There are three resistors connected in parallel, the resistance of each resistor is 3 ohm. What is the total resistance of all the three resistors? [1]
a) 3 ohm
b) 6 ohm
c) 15 ohm
d) 1 ohm

35. Which among the following is the correct way connecting ammeter and voltmeter in the circuit to determine the equivalent resistance of two resistors in series? [1]

a) Both ammeter and voltmeter in parallel

b) Ammeter in series and voltmeter in parallel

c) Ammeter in parallel and voltmeter in series

d) Both ammeter and voltmeter in series

36. A wire that has resistance R is cut into two equal pieces. The two parts are joined in parallel. What is the resistance of the combination? [1]

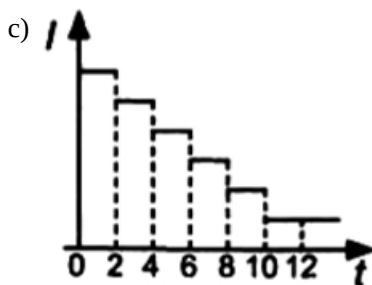
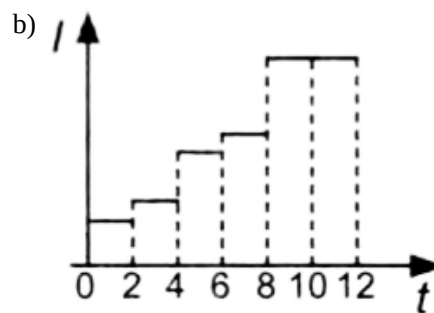
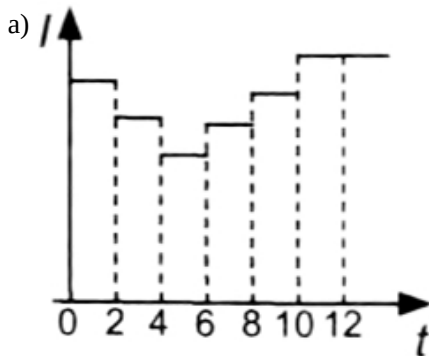
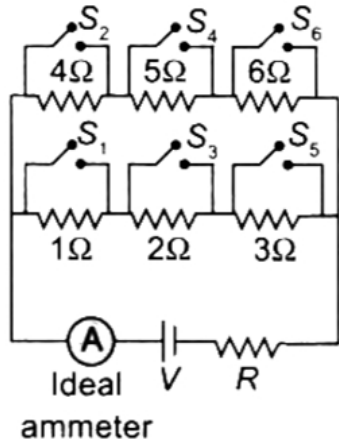
a) $\frac{R}{2}$

b) $\frac{R}{6}$

c) $\frac{R}{4}$

d) None of these

37. Say switches S_1 , S_2 and so on up to S_6 are closed at regular intervals of 2 minutes starting from $t = 0$. The graph of current I (in ampere) versus time t (in minute) is best represented as [1]



38. If a person has five resistors each of value $\frac{1}{5}\Omega$, then the maximum resistance he can obtain by connecting them is [1]

a) 25Ω

b) 1Ω

c) 5Ω

d) 10Ω

39. The lamps in a household circuit are connected in parallel because: [1]

a) this way they required less current

b) if one lamp fails the others also fail

c) if one lamp fails the others remain lit

d) this way they require less power

40. Two LED bulbs of 10W and 5W are connected in series. If the current flowing through 5W bulb is 0.005A, the current flowing through 10W bulb is: [1]

a) 0.0025A

b) 0.005A

c) 0.02A

d) 0.01A

41. A $4\ \Omega$ coil and a $2\ \Omega$ coil are connected in parallel. What is their combined resistance? A total current of 3A passes through the coils. What current passes through the $2\ \Omega$ coil? [1]

a) 4 A

b) None of these

c) 2 A

d) 1 A

42. What possible values of resultant resistance one can get by combining two resistances, one of value 2 ohm and the other 6 ohm? [1]

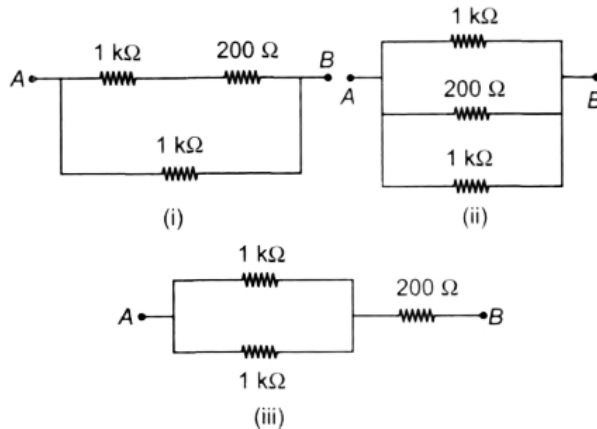
a) 8 ohm

b) 5 ohm

c) 10 ohm

d) None of these

43. In which of the following network of resistors the equivalent resistance between points A and B is highest? [1]



a) All have equal equivalent resistance.

b) Network (iii)

c) Network (i)

d) Network (ii)

44. If four identical resistors, of resistance 8 ohm, are first connected in series so as to give an effective resistance R_s , and then connected in parallel so as to give an effective resistance R_p , then the ratio $\frac{R_s}{R_p}$ is [1]

a) 16

b) 32

c) 2

d) 0.5

45. Two resistors of resistance $2\ \Omega$ and $4\ \Omega$ when connected to a battery will have [1]

a) same current flowing through them when connected in series

b) same potential difference across them when connected in series

c) different potential difference across them when connected in parallel

d) same current flowing through them when connected in parallel

46. The potential difference in a circuit in which components are connected in parallel: [1]

a) Gets divided across each component

b) Potential difference does not appear

c) Gets distributed equally

d) Remains the same across each component

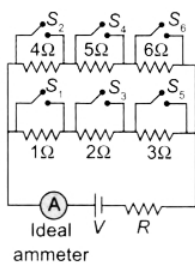
47. R_1 and R_2 are two resistors and r_1 and r_2 are equivalent resistances in series and parallel respectively, then $\frac{R_1}{R_2} =$ [1]

a) $\frac{r_1 r_2}{r_1 + r_2}$

b) $\frac{r_1 + r_2}{r_1 r_2}$

c) $\frac{r_1 + \sqrt{r_1^2 - 4r_1 r_2}}{r_1 - \sqrt{r_1^2 - 4r_1 r_2}}$

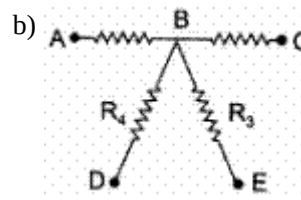
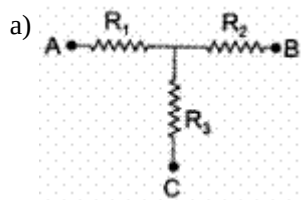
d) $\frac{r_1 + \sqrt{r_1^2 - 4r_1 r_2}}{r_1 + \sqrt{r_1^2 + 4r_1 r_2}}$



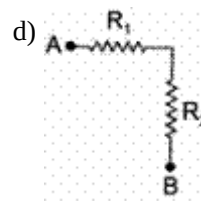
- a) S_2, S_4, S_5 closed
 b) S_1, S_2, S_3 are closed
 c) S_1, S_3, S_5 are closed
 d) S_2, S_3, S_4 closed

64. The combination exhibiting series combination of resistors is

[1]



c) All of these



65. Two LED bulbs of 12W and 6W are connected in series. If the current through 12W bulb is 0.06A the current through 6W bulb will be:

[1]

- a) 0.12A
 b) 0.04A
 c) 0.08A
 d) 0.06A

66. What is the maximum resistance which can be made using five resistors each of $\frac{1}{5} \Omega$?

[1]

- a) 1Ω
 b) $\frac{1}{5} \Omega$
 c) 10Ω
 d) 5Ω

67. What is the minimum resistance which can be made using five resistors each of $\frac{1}{5} \Omega$?

[1]

- a) $\frac{1}{25} \Omega$
 b) 25Ω
 c) $\frac{1}{10} \Omega$
 d) $\frac{1}{5} \Omega$

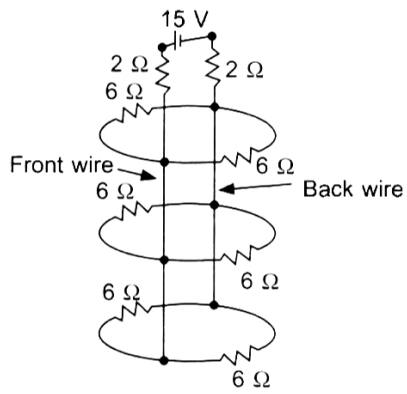
68. If two resistors of 25Ω and 15Ω are joined together in series and then placed in parallel with a 40Ω resistor, the effective resistance of the combination is:

[1]

- a) 0.1Ω
 b) 10Ω
 c) 20Ω
 d) 40Ω

69. Find the current supplied by the source in figure. The resistors are mounted in a cylindrical form.

[1]



- a) $\frac{14}{13} A$ b) 2 A
 c) 3 A d) 1.75 A

70. Three resistances of $2\ \Omega$, $3\ \Omega$ and $5\ \Omega$ are connected in parallel to a 10 V battery of negligible internal resistance. The potential difference across the three resistances will be:

[1]

- a) 5 V b) 10 V
 c) 3 V d) 2 V